

KIERAN PARANJPE

VANCOUVER, BC & MONTREAL, QC

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EDUCATION

McGill University

BSc in Computer Science, Minor in Cognitive Science | 3.98 GPA

2027

Montreal, QC

SKILLS

Languages: C#, Java, Python, JavaScript, TypeScript, SQL, Golang, Bash, HTML, CSS

Frameworks/Tools: Git, Spring Boot, Docker, PyTorch, React.js, Express.js, Unity, Linux, Firebase, Splunk

Soft Skills: Public Speaking, Leadership, Concise Communication, Quick Learning, Teamwork

EXPERIENCE

Software Engineering Intern | Java, TypeScript, Spring Boot, Git, Docker, AGILE, Splunk, JUnit

Autodesk

January 2025 - August 2025

Montreal, QC

- Implemented a **cycle detection algorithm** to ensure batches of commands can be topologically sorted by combining **depth first search** and a **greedy solution to the 'hitting set problem'**, resulting in **30% fewer commands** generated and a **20% speedup**.
- Implemented a **REST client** for search by creating data models and handling errors, achieving **100% test coverage**.
- Added polymorphic-type filter support for search by calling our Types REST API to expand a single RSQL operator into multiple RSQL operators, maintaining 100% test coverage.
- Created an **end-to-end deploy test suite** for the search service containing **over 50 tests** to run **locally and in Jenkins** by syncing test data between local and staging, managing **run-time SQL injection** and writing **parameterised JUnit tests**.
- Configured a **new Nginx Docker container** as a reverse proxy for routing in local search tests.
- Contributed to **critical search features** (listed above), shipped in **5 client deliverables over 4 months**.
- Debugged an issue where purged data was returned to clients by **tracing Splunk logs** as part of my duties **on support**.
- Added support for force purging data on the client and server by introducing a flag, resulting in a **400% performance increase**.

Software Engineering Intern | Python, PyTorch, C#, Unity, JavaScript

The Verse

May 2024 - August 2024

Vancouver, BC / Remote

- Developed a library that tracks breath rate in real-time using microphone input by training a **convolutional neural network** that takes mel spectrograms as input with **PyTorch**, achieving classification accuracy of 85%.
- Created a breath audio dataset with over 50 minutes of annotated breathing samples by implementing a web-app made with **JavaScript and p5.js** that records breath audio and uploads it to a **Firebase storage bucket**.
- Ported and **optimised the PyTorch model to run in C#** to be used in Unity, yielding a **5x speedup** by converting the model to .ONNX, and analysing the running time of specific functions using the **Unity profiler**.
- Reverse engineered PyTorch's short time Fourier transform, spectrogram, and mel spectrogram by stepping through PyTorch source code with a debugger and reproducing its functionality in C#.

Lead Software Engineer, Lead Robot Designer | Java, OpenCV, Android Studio, CAD

FIRST Robotics (FIRST Tech Challenge & FIRST Global Challenge)

September 2019 - April 2023

Vancouver, BC

- Captained my FIRST Tech Challenge robotics team to **1 world-championship qualification** and multiple top 3 provincial finishes. Selected to represent **Team Canada** for the 2022 FIRST Global Challenge in Geneva.
- Enhanced autonomous performance by applying **computer vision** techniques like AprilTag detection and colour masking.
- Implemented **odometry localisation** by combining encoder sensor data from dead wheels and measurements from an IMU.

PROJECTS

Custom Neural Network | Python, NumPy

June 2024

- Programmed a feedforward neural network in Python using NumPy **without any machine learning libraries** (like PyTorch) to classify the MNIST digit dataset with **95% accuracy**.
- Implemented a deep learning network with over 25,000 trainable parameters by researching the underlying linear algebra and calculus behind **back propagation** and **gradient descent**.

URL Shortener | Golang, TypeScript, Next.js, PostgreSQL, Docker, AWS EC2

May 2024

- Developed a full-stack web application to shorten URLs using a Golang server that interacts with a **PostgreSQL** database, running on an AWS EC2 instance with a Next.js frontend.
- Implemented middleware that handles **JSON Web Tokens (JWT)** to ensure users are properly authenticated.
- Encapsulated the backend in a **Docker** container to allow for easy deployment on **AWS EC2**.

KP Games (multiple projects) | C#, Unity, JavaScript, Firebase, 3D Math, Blender

September 2016 - August 2023

- Published 12 video games over 7 years** on itch.io and Google Play using Unity and C#, garnering **over 1000 total users**.
- Developed an active ragdoll platforming game by applying Unity's **physics engine** to map **rigged animations** onto physical joints.
- Implemented **finite state machines** and **behaviour trees** alongside Unity's **NavMesh** across projects to bolster NPC intelligence.
- Designed and created a **multiplayer first person shooter** using Photon Unity Networking, including support for **matchmaking**, team game modes and free for all, automatic respawns, **synchronised movement**, shooting, and powerups.